

# Clos Network Architecture Design

## Streamlined Lab Exercise (90 minutes)

**Duration:** 90 minutes

**Prerequisites:** Network fundamentals, switching concepts, data center basics

## Learning Objectives

Upon completion of this lab, students will be able to:

- Compare 2-tier and 3-tier Clos network architectures
- Calculate switch requirements and bandwidth for data center networks
- Evaluate oversubscription ratios and their impact on performance
- Select appropriate architecture based on scale and workload requirements

## Background

Clos networks provide scalable, fault-tolerant architectures for modern data centers. The 2-tier (spine-leaf) design connects all leaf switches to all spine switches, providing predictable two-hop latency. The 3-tier (super spine-spine-leaf) design adds a super spine layer to scale beyond 2-tier port density limitations, organizing switches into pods for hyperscale deployments.

**Key Concept - Oversubscription Ratio:** The ratio of server bandwidth to uplink bandwidth. A 3:1 ratio means three servers share one unit of uplink capacity. Lower ratios (like 1:1 non-blocking) provide better performance but cost more.

## Exercise Overview

This lab uses the Clos Network Configuration Tool to design networks for two scenarios: a small enterprise deployment and an AI training cluster. You will explore architecture decisions, calculate requirements, and compare design trade-offs. The tool can be accessed at <http://209.38.51.129>

## Part 1: Enterprise Data Center (40 minutes)

### Scenario

Deploy a private cloud supporting 256 servers with 50% growth capacity. Budget allows quality equipment but requires cost justification.

### Requirements

- Initial: 256 servers, Growth: 384 servers

- Leaf switches: 48-port, Uplink: 100 Gbps, Workload: Mixed applications

### Task 1.1: Design with 3:1 Oversubscription (10 min)

Configure the tool with these parameters and record results:

- Servers: 256, Leaf switches: 8, Ports/leaf: 48, Oversubscription: 3:1, Spine switches: 4

| Metric                  | Value |
|-------------------------|-------|
| Servers per leaf        |       |
| Downlink ports per leaf |       |
| Uplink ports per leaf   |       |
| Total switches          |       |

### Task 1.2: Compare with 1:1 Non-Blocking (10 min)

Change oversubscription to 1:1 and adjust spine count. Record:

| Metric                     | Value |
|----------------------------|-------|
| Uplink ports per leaf      |       |
| Number of spine switches   |       |
| Additional switches vs 3:1 |       |

### Task 1.3: Growth Analysis (5 min)

Verify the 3:1 design supports 384 servers (48 per leaf). Calculate if this fits in 48 ports.

### Analysis Questions (15 min)

#### Q1: Calculate cost difference

Spine switches: \$50,000 each, 100G optics: \$500 per link. Show calculations for 3:1 vs 1:1 designs.

#### Q2: Impact of spine failure

In 3:1 design, what happens when one spine fails? Calculate new oversubscription ratio and bandwidth loss percentage.

**Q3: Design recommendation**

Which design (3:1 or 1:1) do you recommend? Consider cost, growth, and application needs.

## Part 2: AI Training Cluster (40 minutes)

### Scenario

Build a GPU cluster for large language model training requiring maximum bandwidth and minimal latency. Initial: 512 servers, planned growth to 1,024 servers.

### Requirements

- Capacity: 512 → 1,024 servers
- Leaf switches: 64-port, Uplink: 400 Gbps
- Performance: 1:1 non-blocking required for AI workloads

### Task 2.1: Initial 2-Tier Design (12 min)

Design for 512 servers using 32 servers per leaf (50% ports for downlinks):

| Calculation/Metric              | Value |
|---------------------------------|-------|
| Leaf switches needed (512 ÷ 32) |       |
| Number of spine switches        |       |
| Ports per spine                 |       |

### Task 2.2: Evaluate 2-Tier Scalability (8 min)

Attempt scaling to 1,024 servers with 2-tier. Assume 128-port spine limit.

| Metric                          | Value |
|---------------------------------|-------|
| Leaf switches for 1,024 servers |       |
| Ports required per spine        |       |
| Is 2-tier feasible? (Yes/No)    |       |

### Task 2.3: Design 3-Tier Architecture (10 min)

Configure 3-tier for 1,024 servers. Experiment with pod structure:

| Parameter              | Value |
|------------------------|-------|
| Number of pods         |       |
| Spines per pod         |       |
| Number of super spines |       |
| Total switches         |       |

## **Analysis Questions (10 min)**

### **Q1: Why non-blocking for AI?**

Explain why AI/ML training requires 1:1 non-blocking design. How does oversubscription affect training time?

### **Q2: Latency comparison**

Calculate maximum hop count in 3-tier vs 2-tier. Assume 500 ns per switch hop. What is the latency difference?

### **Q3: Pod architecture benefits**

Explain how pods provide operational benefits for large-scale deployments. Consider maintenance and failure isolation.

## Part 3: Synthesis and Key Takeaways (10 minutes)

### Architecture Selection Framework

Based on your designs, complete this decision framework:

| Criterion              | 2-Tier | 3-Tier |
|------------------------|--------|--------|
| Server count range     |        |        |
| Best for workload type |        |        |
| Maximum hop count      |        |        |
| Complexity             |        |        |

### Key Learning Points

Summarize the most important insights from this lab:

- When to choose 2-tier vs 3-tier architecture:
- Impact of oversubscription on cost vs performance:
- Why AI workloads require different designs than general applications:

## Submission Requirements

Submit this completed worksheet with:

- All data tables filled in
- Complete answers to all analysis questions
- Screenshots from the Clos Configuration Tool showing your final designs
- Completed synthesis framework and key takeaways